

ARC Figures

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```
library(tidyverse)
library(dplyr)
library(readxl)
library(janitor)
library(ggplot2)
library(ggtext)
library(stringr)
library(patchwork)
```

```
frequencies_renamed <- read_excel("Frequencies of PI and Cxn to CPS.xlsx")
frequencies_renamed <- clean_names(frequencies_renamed) |>
  mutate(denominator = case_match(denominator,
                                   "imprisoned people" ~ "Imprisoned People",
                                   "incarcerated parents" ~ "Incarcerated Parents",
                                   "foster care children" ~ "Foster Care Children",
                                   "CPS-involved children" ~ "CPS Involved-Children"
                                   ))
```

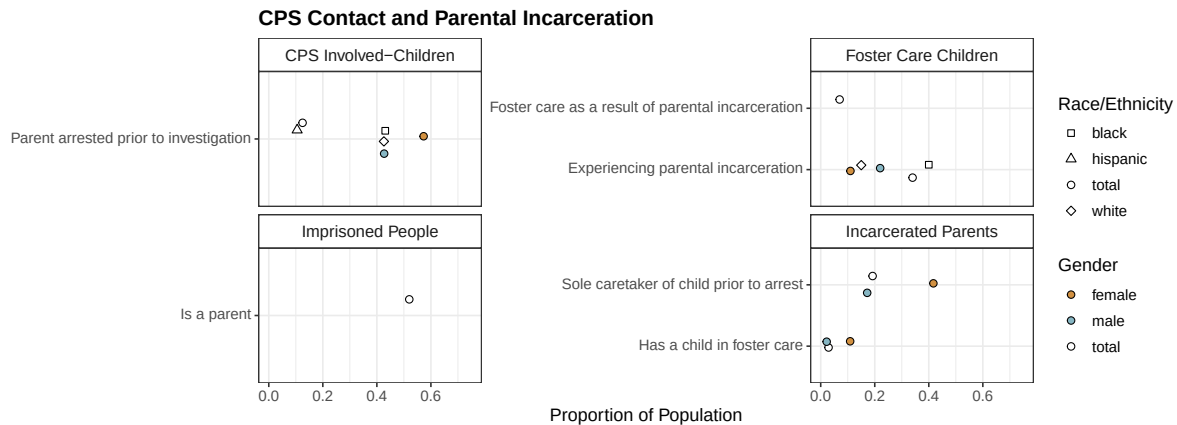
```
frequencies_pivoted <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, white, black, hispanic),
               names_to = "demographic",
               values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("white", "black", "hispanic") ~ demographic,
    TRUE ~ "total"
  ),
  gender = case_when(demographic %in% c("male", "female") ~ demographic,
                     TRUE ~ "total"),
  cps_cj = if_else(denominator %in% c("Incarcerated Parents",
                                       "Imprisoned People"),
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        "Criminal Justice Contact",
        "Child Protective Services Contact")) |>
select(-demographic)

frequencies_pivoted |>
  ggplot(aes(x = proportion, y = clean_name,
            fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15, width = 0),
            color = "black",
            show.legend = TRUE) +
  facet_wrap(~denominator, scales = "free_y") +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
                              "total" = "white")) +
  scale_shape_manual(values = c("black" = 22, "hispanic" = 24,
                              "white" = 23, "total" = 21)) +
  theme_bw() +
  theme(strip.background = element_rect(fill = NA),
        axis.title.y = element_blank(),
        strip.text.x = element_text(
          face = "plain",
          size = 11,
          margin = margin(t = 6, b = 6)
        ),
        plot.title = element_text(face = "bold"),
        text = element_text(size = 12),
        axis.text.y = element_markdown(),
        plot.caption.position = "plot",
        plot.caption = element_markdown(hjust = 0)) +
  guides(fill = guide_legend(
    override.aes = list(shape = 21, color = "black"))) +
  xlim(0, 0.75) +
  labs(title = "CPS Contact and Parental Incarceration",
       x = "Proportion of Population",
       y = "Condition",
       shape = "Race/Ethnicity",
       fill = "Gender")

```



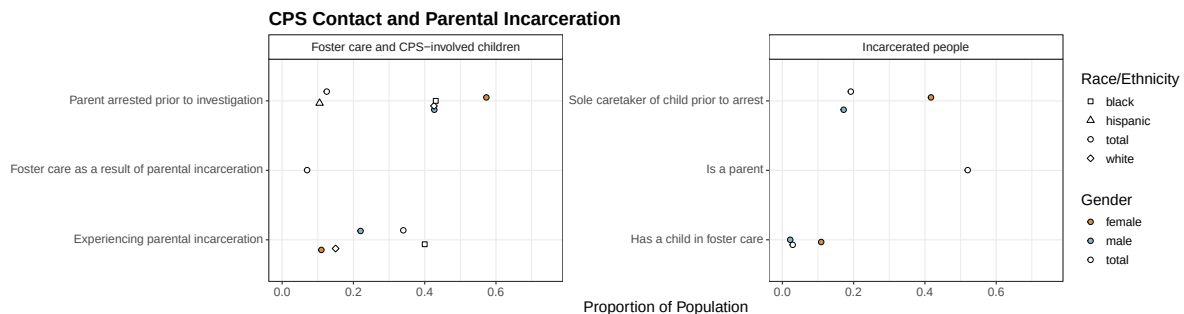
```
frequencies_mutated <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, white, black, hispanic),
    names_to = "demographic",
    values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("white", "black", "hispanic") ~ demographic,
    TRUE ~ "total"),
  gender = case_when(demographic %in% c("male", "female") ~ demographic,
    TRUE ~ "total"),
  alt_denominator = case_when(
    str_detect(clean_name, "parental incarceration") ~ "Foster care and CPS-involved children",
    str_detect(clean_name, "Parental incarceration") ~ "Foster care and CPS-involved children",
    str_detect(clean_name, "Parent arrested") ~ "Foster care and CPS-involved children",
    TRUE ~ "Incarcerated people"),
  ) |>
  select(-demographic)

frequencies_mutated |>
  ggplot(aes(x = proportion, y = clean_name,
    fill = gender, shape = race)) +
  geom_point(size = 2,
    position = position_jitter(height = .15, width = 0),
    color = "black",
    show.legend = TRUE) +
  facet_wrap(~alt_denominator, scales = "free_y") +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
    "total" = "white")) +
  scale_shape_manual(values = c("black" = 22, "hispanic" = 24,
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"white" = 23, "total" = 21)) +
theme_bw() +
theme(strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      plot.title = element_text(face = "bold"),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
guides(fill = guide_legend(
  override.aes = list(shape = 21, color = "black"))) +
xlim(0, 0.75) +
labs(title = "CPS Contact and Parental Incarceration",
     x = "Proportion of Population",
     y = "Condition",
     shape = "Race/Ethnicity",
     fill = "Gender")

```



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frequencies_mutated2 <- frequencies_mutated |>
  filter(denominator == "CPS Involved-Children")

p3 <- frequencies_mutated2 |>
  ggplot(aes(y = clean_name, x = proportion,
             fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15, width = 0),
            color = "Black",

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      show.legend = TRUE) +
scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
                             "total" = "White")) +
scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,
                              "White" = 23, "total" = 21)) +
scale_y_discrete(labels = c("Parent arrested prior to investigation" =
                             "Parent arrested prior<br>to investigation"
                              )) +

theme_bw() +
theme(legend.position = "none") +
theme(strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      axis.title.x = element_text(size = 10),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      legend.title = element_text(size = 10),
      plot.title = element_text(face = "bold",
                                size = 12),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
xlim(0, 0.75) +
labs(title = "CPS-Involved Children",
     x = "% of CPS-Involved Children",
     y = "Condition",
     shape = "Race/Ethnicity",
     fill = "Gender")

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frequencies_mutated3 <- frequencies_mutated |>
  filter(denominator == "Foster Care Children")

p4 <- frequencies_mutated3 |>
  ggplot(aes(y = clean_name, x = proportion,
             fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15, width = 0),
            color = "Black",
            show.legend = TRUE) +

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scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
                             "total" = "White")) +
scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,
                              "White" = 23, "total" = 21)) +
scale_y_discrete(labels = c("Experiencing parental incarceration" =
                             "Experiencing parental<br>incarceration",
                             "Foster care as a result of parental incarceration" =
                             "Foster care as a<br>result of parental<br>incarceration"
                             )) +

theme_bw() +
theme(legend.position = "none",
      strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      axis.title.x = element_text(size = 10),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      plot.title = element_text(face = "bold",
                                size = 12),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
xlim(0, 0.75) +
labs(x = "% of Foster Care Children",
     y = "Condition",
     shape = "Race/Ethnicity",
     fill = "Gender")

```

```

frequencies_mutated4 <- frequencies_mutated |>
  filter(denominator == "Imprisoned People")

p5 <- frequencies_mutated4 |>
  ggplot(aes(y = clean_name, x = proportion,
             fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15, width = 0),
            color = "Black",
            show.legend = TRUE) +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",

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        "total" = "White")) +
scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,
        "White" = 23, "total" = 21)) +
scale_y_discrete(labels = c("Is a parent" =
        "Is a<br>parent"
        )) +
theme_bw() +
theme(legend.position = "none") +
theme(strip.background = element_rect(fill = NA),
        axis.title.y = element_blank(),
        axis.title.x = element_text(size = 10),
        strip.text.x = element_text(
                face = "plain",
                size = 11,
                margin = margin(t = 6, b = 6)
        ),
        plot.title = element_text(face = "bold",
        size = 12),
        text = element_text(size = 14),
        axis.text.y = element_markdown(),
        plot.caption.position = "plot",
        plot.caption = element_markdown(hjust = 0)) +
xlim(0, 0.75) +
labs(title = "Incarcerated Adults",
        x = "% of Imprisoned Adults",
        y = "Condition",
        shape = "Race/Ethnicity",
        fill = "Gender")

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frequencies_mutated5 <- frequencies_mutated |>
  filter(denominator == "Incarcerated Parents")

p6 <- frequencies_mutated5 |>
  ggplot(aes(y = clean_name, x = proportion,
        fill = gender, shape = race)) +
  geom_point(size = 2,
        position = position_jitter(height = .15, width = 0),
        color = "Black",
        show.legend = TRUE) +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
        "total" = "White")) +
  scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,

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      "White" = 23, "total" = 21)) +
scale_y_discrete(labels = c("Sole caretaker of child prior to arrest" =
      "Sole caretaker of<br>child prior to <br>arrest",
      "Has a child in foster care" =
      "Has a child<br>in foster care")) +
theme_bw() +
theme(strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      axis.title.x = element_text(size = 10),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
guides(fill = guide_legend(
  override.aes = list(shape = 21, color = "black"))) +
xlim(0, 0.75) +
labs(x = "% of Imprisoned Parents",
     y = "Condition",
     shape = "Race/Ethnicity",
     fill = "Gender")

(p3 + p5) / (p4 + p6)

```

Warning: Removed 3 rows containing missing values or values outside the scale range (``geom_point()``).

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).

